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Development of performance analysis model with numerical and dynamic energy modeling for an integrated PVT-HP system

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ABSTRACT

This study delivers a comprehensive numerical analysis of an integrated photovoltaic/thermal-heat pump (PVT-HP) system, underscored by an innovative three-dimensional thermal energy storage (TES) model. Addressing the limitations of conventional one-dimensional TES models, which struggled with accurately simulating horizontal flows and the fluid state at various facility inlets, this research marks a significant advancement. Our development of a 3D TES model optimizes finite elements to reduce computation time and enhance accuracy, vital for precise energy analysis in buildings and air conditioning systems. The integration of this model with a network model allows for detailed analysis of flow rate changes and their impacts on heat transfer, crucial for the integrated PVT-HP system. Key findings include a 30 % increase in the temperature difference between the top and bottom layers of the TES when the flow rate is reduced from 50 LPM to 20 LPM, and a 1 % decrease in the heat pump's coefficient of performance (COP) with a 10 LPM reduction in flow rate. The study emphasizes that managing the flow rate is key to optimizing temperature gradients and system efficiency. Strategic adjustments in flow rate significantly affect thermal storage and transfer capabilities, proving essential in system performance. This research offers vital insights into optimizing PVT-HP systems, focusing on the importance of flow rate control in improving temperature management and overall system performance.

1. Introduction

PV (Photovolatic)'s direct conversion of solar radiation into electricity has been spotlighted as an alternative to traditional fossil fuel sources (Deshwal et al., 2021; Le, 2023; Tawalbeh et al., 2021). However, PV systems inherently suffer from vulnerabilities to ambient temperature variations, typically leading to decreased efficiency with rising temperatures (Dhassa et al., 2014). To mitigate this thermal bottleneck, cooling fans have been employed (Praveenkumar et al., 2022), but this approach has not been widely adopted due to its low thermal dissipation efficiency and the additional electrical energy required for heat discharge.

To address these limitations, the concept of photovoltaic-thermal (PVT) systems was introduced. PVT not only generates electricity but also captures waste heat from PV cells during the cooling process for practical applications such as heating and hot water supply, offering a more comprehensive and sustainable approach to solar energy utilization (Dwivedi et al., 2020; Elbreki et al., 2017).

Despite these advancements, the mismatch between the supply of renewable energy and consumer demand times has continued to pose challenges for the efficient use of renewable energy systems. This has led to the emergence of thermal energy storage (TES) as a key research area, integrating TES into renewable energy infrastructures (Chae et al., 2023a). In this context, the development of the multi-source thermal energy storage (MSTES) system, which utilizes solar energy via PVT and ambient air heat via heat pumps (HP), has been realized.

The MSTES system based on PVT-HP incorporates the transfer of heat energy collected from the PVT and generated by the HP through circulating water to the TES. This stored thermal energy is then conveyed through circulating water to fan coil units (FCU) for indoor heating or cooling. The MSTES system addresses the inherent imbalance between supply and demand in renewable energy facilities and allows for heat accumulation using the HP under favorable conditions, enhancing system performance and reducing the required heat production by the HP during peak loads (Zhu et al., 2014).

Recent studies in this field have highlighted the importance of developing optimal control methods or capacity design to enhance the economic viability of MSTES systems. Sazon and Nikpey (2024) confirmed through parametric research and dynamic simulation that TES can influence the temperature and pressure of heat pump refrigerants, thereby improving the seasonal performance factor (SPF) and

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Nomenclatures

A	area (m^2)
$C_{p,w}$	specific heat of water ($J/(kg \bullet m^2)$)
F'	heat transfer efficiency adjustment factor
H	pump head (m); hour angle (rad)
I	solar radiation (W/m^2)
L	length (m)
P	power consumption (W); power generation (W); pressure (kPa)
R	aborbed resistance ($m^2 \bullet K/W$); thermal resistance ($m^2 \bullet K/W$)
S	net absorbed solar radiation (W/m^2)
T	temperature (℃)
W	distance (m)
b	energy balance temperature factor (W/m^2)
d	diameter (m)
g	acceleration of gravity (m/s^2)
h	loss coefficient ($W/(m^2 \bullet K)$); convective heat transfer coefficient ($W/(m^2 \bullet K)$); enthalpy (J/kg)
j	thermal resistance adjust factor ($W/(m^2 \bullet K)$)
k	thermal conductivity ($W/(m \bullet K)$)
m	flow rate (kg/s); fin efficiency calculation parameter
n	conductor heat transfer coefficient ($W/(m \bullet K)$)
q	heat transfer (W)
q'	heat flux (W/m)
t	time sequence (s)
u	tube ehat transfer product ($W/(m \bullet K)$)
z	thermal resistance in PVT ($m \bullet K/W$)

Greek symbols

$(\tau\alpha)_n$	normal incidence transmittance
α	solar altitude angle (rad)
β	module tilt from horizontal (rad); temperature coefficient of efficiency (K^{-1})
γ	PV efficiency coefficient
δ	declination angle (rad)
ε	temperature rise efficiency factor (W/m)
ε_s	Skyward long-wave emissivity
η	efficiency (%)
θ	thermal transfer effectiveness ratio
λ	thickness (m)
σ	net heat transfer coefficient ($W/(m \bullet K)$)
σ_s	Stefan-Boltzmann constant ($W/(m^2 \bullet K^4)$)
φ	latitude (rad)
ψ	module azimuth (rad)
ϕ	solar azimuth angle (rad)

Acronyms

ASHP	air source heat pump
CFD	computational fluid dynamics
COP	coefficient of performance
FCU	fan coil units
GSHP	ground source heat pump
HP	heat pump
LMTD	log mean temperature difference (℃)
LPM	liter per minute
MLP	multilayer perceptron
MSE	mean square error
MSTES	multi-source thermal energy storage
NOCT	nominal operating cell temperature (℃)
P&ID	piping and represents instrumentation diagram
PSO	particle swarm optimization
PV	photovoltaic
PVT	photovoltaic/thermal
SPF	seasonal performance factor
TES	thermal energy storage
sMAPE	symmetric mean absolute percentage error (%)

the levelized cost of heating (LCOH). Similarly, researchers like Cho et al. (2021) and Chae et al. (2023b) have demonstrated significant performance improvements through the control of circulating water flow alone. Moreover, studies by Elaouzy and El Fadar (2023) and Bae et al. (2023) simulations have quantitatively verified the positive environmental impact and profitability enhancement through increased renewable energy utilization rates using simulations. Additionally, Obalanlege et al. (2020) emphasized the need for optimizing the capacity and fluid flow methods in MSTES systems. These studies showcase that, despite the utilization of renewable energy facilities, economically favorable outcomes are achievable with appropriate system design and operation.

Despite confirming the economic and environmental benefits of MSTES systems, their application in actual buildings remains rare. This scarcity is attributed to the complexity of fluid dynamics within TES systems, especially when internal heat exchangers are involved. To prevent pipeline freeze damage during cold seasons, antifreeze is mixed with the circulating water in PVT systems. This mixed antifreeze is indirectly exchanged through a coil-shaped heat exchanger within the TES to prevent mixing with other systems. Previous research simplified the TES numerical model to one dimension for energy analysis, complicating the consideration of lateral heat transfer processes and hampering accurate predictions of indirect heat exchange amounts transferred from the coil-shaped heat exchanger. Consequently, the practical application of simulation-derived optimal capacities and developed control logic has been challenging (Langerova and Matuska, 2023; Osterman and Stritih, 2021).

In the field of mechanical engineering, computational fluid dynamics (CFD) tools have been employed for precise analysis of heat transfer between TES model elements, offering accurate predictions of stratification levels and outlet temperatures based on inlet/outlet locations and heat exchanger positioning (Luo et al., 2023). Research by Abdelhak Mhiri et al. (2015) and Gómez et al. (2018) confirmed the high accuracy of three-dimensional TES models in predicting internal temperature dynamics. As such, the limitations of one-dimensional TES models and the advantages of three-dimensional models are clear. The three-dimensional TES model presents significant advancements over conventional one-dimensional models. Traditional 1D models often struggle to simulate horizontal flows and the fluid state at various facility inlets accurately, leading to limitations in predicting the thermal stratification and energy storage dynamics accurately. Fig. 1 shows how energy and momentum are transferred between nodes between the inlet and outlet depending on the direction of fluid flow for each model.

However, models designed using CFD tools face challenges with computational time and integration with other programs, limiting their application in energy analysis fields with complex, dynamically changing conditions like occupant scenarios and weather conditions.

Therefore, this study developed a three-dimensional TES model based on the finite volume method for the PVT-HP based MSTES system, addressing issues of computational time and accuracy. The 3D TES model developed in this research addresses these limitations by optimizing finite elements to reduce computation time while enhancing accuracy. This model accurately reflects the complexities of thermal energy storage and utilization, enabling precise predictions of

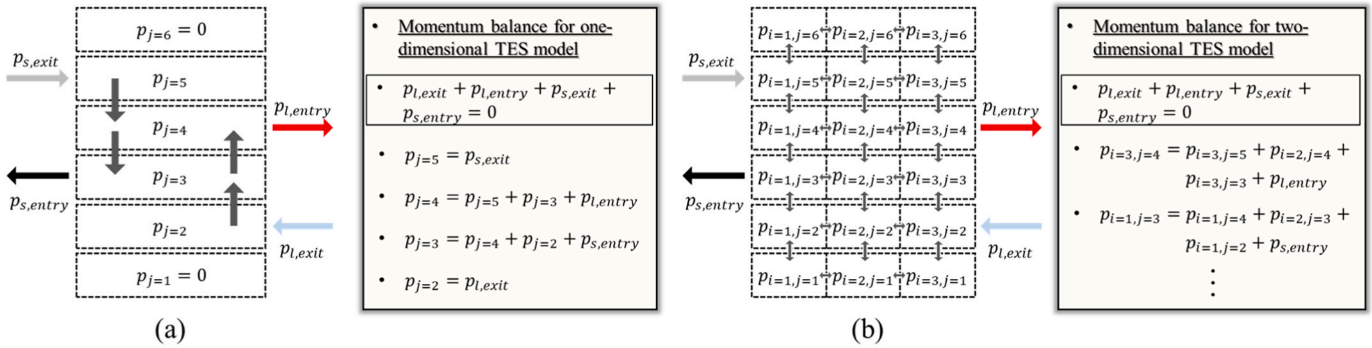

Fig. 1. Comparative flow analysis in TES models: (a) one-dimensional TES model, and (b) multidimensional TES model. $p_{s,exit}$: momentum transferred from the heat source; $p_{s,entry}$: momentum transferred to the heat source; $p_{l,exit}$: momentum transferred from the load side; $p_{l,entry}$: momentum transferred to the load side; i: horizontal-coordinate; and j: vertical-coordinate.

circulating water temperatures at each facility's inlet, facilitating the development of realistic control logic, and allowing for the determination of feasible system capacities.

Ultimately, this research proposes a new approach to enhancing the efficiency of thermal energy storage and usage within building energy systems. By facilitating the integration of renewable energy and the efficient use of multiple heat sources, it significantly contributes to achieving sustainable development goals. Furthermore, it provides essential tools and guidelines for the efficient construction and management of sustainable energy systems, offering new research directions for future scholars.

2. Methodology

2.1. System overview

The focus of this study, the integrated PVT-HP system, is installed in a real building in Busan, Korea, with specific configurations and environmental conditions that may limit the broader applicability of the findings. The essential components include PVT modules, TES, a heat pump, and a circulation pump. The data and results are primarily relevant to this specific setup and location. Fig. 2 presents the Piping and Instrumentation Diagram (P&ID) of the system. Additionally, data on the building's location is summarized in Table 1.

Fig. 3 showcases the components of the PVT system and photographs of the experimental site. The satellite map in this figure is based on public data from the Korea National Geographic Information and Geography Institute (National Land Information Map, 2021). In the PVT setup, PV cells generate electricity through the photoelectric effect, while a fluid flowing through a tube attached to an absorber plate beneath the PV cells collects heat. The TES system features a vertical cylindrical water-filled tank, which includes an immersed heat exchanger but lacks a nozzle connected to the piping. The HP meets heating demands by transferring thermal energy from the storage tank using both a shell and coil system and compression work. Detailed design information about the integrated PVT-HP system is available in Table 2.

Table 1
Geospatial and temporal parameters summary.

Parameter	Value
Latitude	35.23°
Longitude	129.23°
Local standard time meridian	135°
Sea level	41 m

2.2. Numerical modeling

The numerical models developed in this study are tailored to the specific configurations and operational parameters of the integrated PVT-HP system installed in Busan. The models include detailed analyses of flow rate adjustments, temperature gradients within the TES, and the efficiency of heat exchangers. The simulations provide insights into optimizing the system's performance within the given environmental conditions.

2.2.1. Circulator pump model

The power consumption of circulator pumps is primarily influenced by three factors: flow rate ($\dot{m}$), head (H_{pump}), and efficiency (η_{pump}). This study introduces a straightforward methodology to calculate the power consumption of the pump, taking these factors into account. We maintain a consistent head measurement in our experimental setup, with the head for all circulator pumps uniformly set at 11 meters. This standardization is essential for ensuring the accuracy of our power consumption estimates. The efficiency of the pump varies with the flow rate, and is determined using a formula derived from the pump's performance data provided by the manufacturer (MAGNA3 25-120, 2021). The

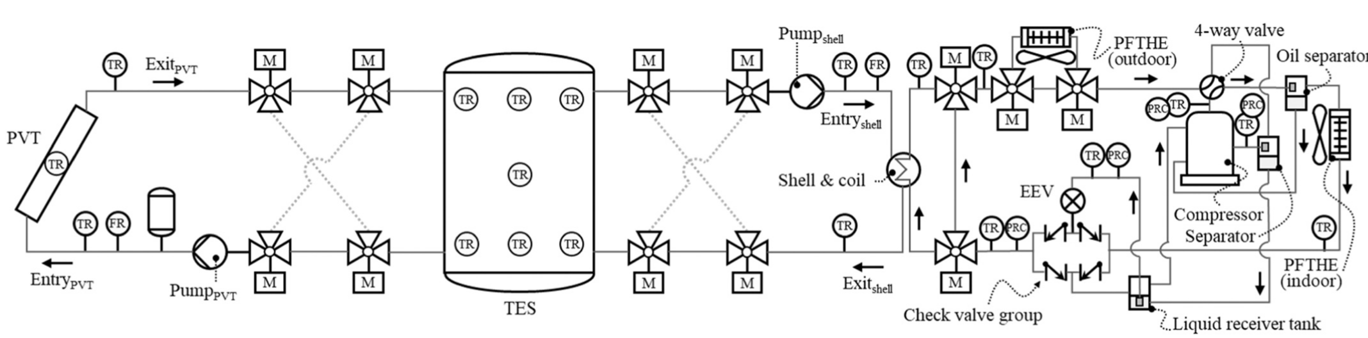

Fig. 2. Schematic of integrated PVT-HP system: piping and instrumentation diagram.

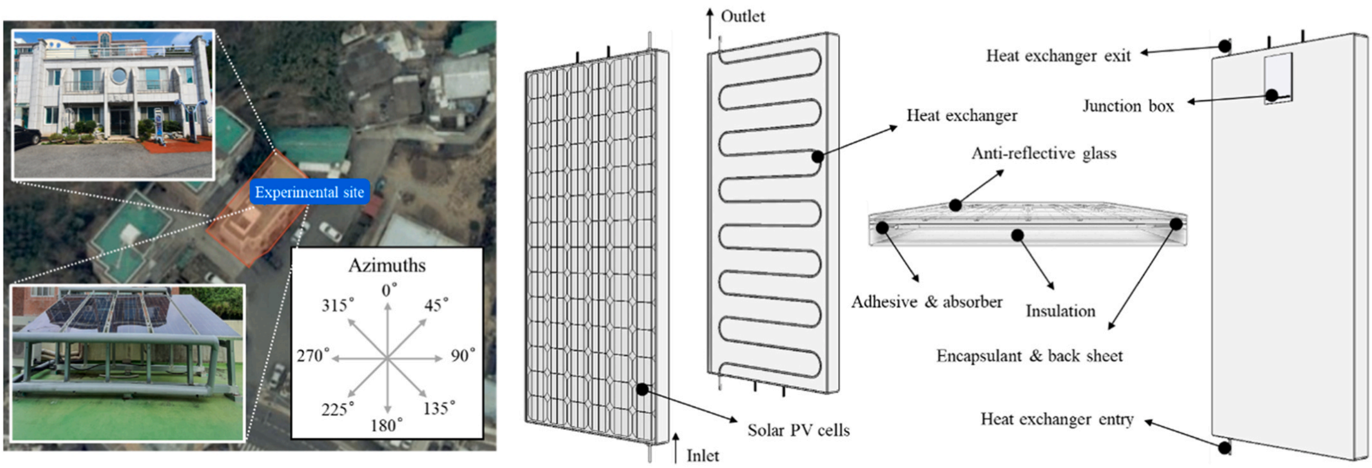

Fig. 3. Annotated schematic of the experimental site and the constituent elements within the PVT.

Table 2
Integrated PVT-HP system: detailed design specifications.

	Parameter		Value
PVT	Number of modules		10
	Number of identical fluid tubes		14
	Length of the collector		2 m
	Width of the collector		1 m
	Tube-to-tube distance		0.14 m
	Absorber thermal conductivity		385 W/(m • K)
	Thickness of the absorber plate		0.0005 m
	Module angle		45°
	Module azimuth		130°
	Tube diameter		0.01 m
	Width of the bond		0.01 m
	Thickness of the bond		0.01 m
	Bond thermal conductivity		377 W/(m • K)
	PV to absorber resistance		0.036 m^2 • K/W
	Backside thermal resistance		10.8 m^2 • K/W
	Normal incidence reflectance		0.15
	Skyward long-wave emissivity		0.9
	Normal incidence transmittance		0.85
	Top convective loss coefficient		5.56 W/(m^2 • K)
	Backside heat transfer coefficient		4.17 W/(m^2 • K)
	Surface emissivity		0.9
TES	Height		1 m
	Volume		0.3 m^3
	Insulation thermal conductivity		0.035 W/(m • K)
	Insulation thickness		0.06 m
	Pipe diameter		0.04 m
Circulator pump	Head		11 m
	Power rating		180 W
Heat pump	Compressor	Type	Scroll
		Refrigerant	R-410a
		Compressor power	3 kW
	Shell and coil	Flow arrangement	Counter flow
		Coil length	7 m
		Coil outside diameter	0.016 m
		Coil inside diameter	0.012 m
		Coil thermal conductivity	398 W/(m^2 • K)
		Coil radius	0.054 m
		Coil pitch	0.025 m
		Coil dimensionless pitch	0.074 m
		Shell diameter	0.12 m
		Shell hydraulic diameter	0.284 m

formula is:

$$\eta_{pump} = 0.2243\ln(\dot{m}) + 0.5336 \tag{1}$$

This empirical equation facilitates the precise calculation of efficiency across various flow rates, which is crucial for accurately predicting the pump's power consumption under different operational conditions. The power consumption (P_{pump}) of the pump is then calculated using the formula:

$$P_{pump} = \frac{\dot{m}H_{pump}g}{\eta_{pump}} \tag{2}$$

Table 3 lists the variables used in calculating the efficiency and power consumption of the circulator pump, along with their corresponding units.

2.2.2. Photovoltaic/thermal model

The PVT system is designed to maximize solar energy utilization by enabling fluid flow between strategically embedded PV cells and tubes connected to absorbers. This innovative design allows for dual functionality: electricity generation through the PV cells and thermal energy collection via the absorbers (Emmanuel et al., 2021). Fig. 4 illustrates the PVT system's architecture, clearly showing both the electricity generation mechanism of the PV cells, and the heat transfer process facilitated by the thermal system components.

The efficacy of power generation and heat collection in PVT systems is significantly influenced by a variety of environmental and system-specific factors. In practical scenarios, particularly in actual building settings, the environmental data typically available includes total solar radiation, outdoor ambient temperature, and relative humidity (Akram and Hasanuzzaman, 2022). However, the limitation in the variety of weather data accessible presents challenges in accurately conducting numerical analyses of the energy performance of PVT systems.

To address these challenges and mitigate the constraints posed by the limited range of environmental sensors, it is essential to enhance the dataset through supplementary data. This study proposes a methodology to estimate the solar radiation incident on the PVT modules. This estimation is based on a combination of regional location and time-specific information, along with total solar radiation data measured by a pyranometer.

One of the factors that has the greatest impact on the performance of the PVT model is the amount of solar radiation applied to the module. Therefore, it is necessary to accurately numerically analyze module solar

Table 3
Nomenclature and values for pump power consumption.

Symbol	Description	Units or value
η_{pump}	Pump efficiency	-
$\dot{m}$	Power generation	kg/s
P_{pump}	Heat collection	W
H_{pump}	Net absorbed solar radiation	11 m
g	PV efficiency	9.81 m/s^2

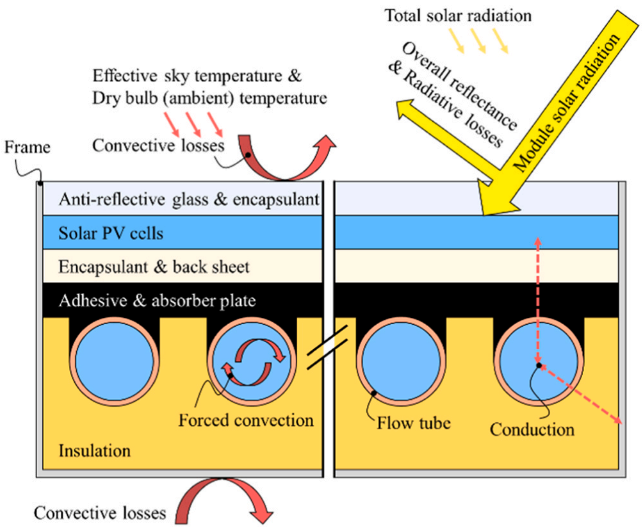

Fig. 4. PVT collector schematic: structural, power generation, and heat transfer features.

radiation from the measured total solar radiation.

Module solar radiation (I_{module}) is calculated along with total solar radiation (I_{total}), solar altitude angle (α), module tilt from horizontal (β), module azimuth (ψ), and solar azimuth angle (ϕ) as follows (Alam Emon et al., 2022):

$$I_{module} = I_{total}(\cos\alpha\sin\beta\cos(\psi-\phi)+\sin\alpha\cos\beta) \tag{3}$$

$$\alpha = \sin^{-1}[\sin\delta\sin\varphi+\cos\delta\cos\varphi\cos(H)] \tag{4}$$

$$\phi = \cos^{-1}\left[\frac{\sin\delta\sin\varphi-\cos\delta\cos\varphi\cos(H)}{\cos\alpha}\right] \tag{5}$$

Table 4 summarizes the variables used in the photovoltaic/thermal model equations for solar radiation calculations, including symbols and units.

Through the calculated module solar radiation, the observed outdoor temperature, and the physical property information of the materials constituting the PVT, the heat collection (q_{PVT}), power generation (P_{PV}), net absorbed solar radiation (S), PV efficiency (η_{PV}), and the radiation heat transfer coefficient (h_{rad}) of the PVT are calculated as follows (Chikate et al., 2015; Combined PV/T Solar Collector, 2013; Duffie et al., 2022; Dwivedi et al., 2020; Luo et al., 2017):

$$q_{PVT} = \dot{m}_{PVT}C_{p,w}\left(T_{PVT,exit}-T_{PVT,entry}\right) \tag{6}$$

$$P_{PV} = (\tau\alpha)_n\eta_{PV}A_{collector}I_{module} \tag{7}$$

$$S = (\tau\alpha)_n I_{module}(1-\eta_{PV}) \tag{8}$$

$$\eta_{PV} = \eta_{PV,ref}\left[1-\beta\left(T_{PV}-T_{PV,ref}\right)+\gamma\log\left(\frac{I_{module}}{I_{ref}}\right)\right] \tag{9}$$

Table 4
Nomenclature and values for pump power consumption.

Symbol	Description	Units or value
I_{module}	Incident solar radiation on module	W/m^2
I_{total}	Total solar radiation	W/m^2
α	Solar altitude angle	radians
β	Module tilt from horizontal	radians
ψ	Module azimuth angle	radians
ϕ	Solar azimuth angle	radians
δ	Declination angle	radians
H	Hour angle	radians

$$h_{rad} = \varepsilon_s\sigma_s\left(T_{PV}+T_{sky}\right)\left(T_{PV}^2+T_{sky}^2\right) \tag{10}$$

$$T_{sky} = T_{amb}-11 \tag{11}$$

Table 5 summarizes the nomenclature and values used in these calculations.

The calculation process for the temperature of the materials that make up the PVT and the fluid within the tube varies depending on whether the circulator pump is operating. When the circulator pump is stopped, the temperature of the fluid in tube is equal to the PV cell temperature, and the PV cell temperature (T_{PV}) is calculated by dividing the operating time sequence (t) as follows:

$$T_{PV} = T_{amb}+(\text{NOCT}-20)\frac{I_{module}}{800}\text{fort}=1T_{PV} = \frac{S(R_t+R_b)+h_{rad}T_{sky}(R_t+R_b)+T_{back}+h_{conv,t}(R_t+R_b)T_{amb}}{1+h_{rad}(R_t+R_b)+h_{conv,t}(R_t+R_b)}\text{fort}\geq 2 \tag{12}$$

Table 6 summarizes the nomenclature and values used in these calculations.

When the circulator pump is operating, the PV cell temperature is calculated as follows:

$$T_{PV} = \frac{D_{tube}\overline{T}_{PV,base}+(W_{tube}-d_{tube})\overline{T}_{PV,fin}}{W_{tube}} \tag{13}$$

where d_{tube} is the tube diameter, $\overline{T}_{PV,base}$ is the base mean temperature of the PV surface, W_{tube} is tube-to-tube distance, and $\overline{T}_{PV,fin}$ is PV fin mean temperature.

$\overline{T}_{PV,base}$ and $\overline{T}_{PV,fin}$ are respectively calculated as follows:

$$\overline{T}_{PV,base} = R_tF'\left(S+h_{rad}T_{sky}+h_{conv,t}T_{amb}+\frac{\overline{T}_{base}}{R_t}\right) \tag{14}$$

$$\overline{T}_{PV,fin} = R_tF'\left(S+h_{rad}T_{sky}+h_{conv,t}T_{amb}+\frac{T_{fin}}{R_t}\right) \tag{15}$$

Here, F' is heat transfer efficiency adjustment factor, $\overline{T}_{base}$ is base mean temperature, and T_{fin} is fin mean temperature.

Calculations for F', $\overline{T}_{base}$, and T_{fin}:

$$F' = \frac{1}{h_{rad}R_t+h_{conv,t}R_t+1} \tag{16}$$

Table 5
Nomenclature and values for PVT system calculations.

Symbol	Description	Units or value
q_{PVT}	Heat collection	W
P_{PV}	Power generation	W
S	Net absorbed solar radiation	W/m^2
η_{PV}	PV efficiency	%
h_{rad}	Radiation heat transfer coefficient	W/(m^2·K)
$\dot{m}_{PVT}$	PVT fluid flow rate	kg/s
$C_{p,w}$	Specific heat of water	4193 J/(kg·K)
$T_{PVT,exit}$	PVT fluid exit temperature	℃
$T_{PVT,entry}$	PVT fluid entry temperature	℃
$(\tau\alpha)_n$	Normal incidence transmittance	0.85
$A_{collector}$	Solar collector top area	m^2
$\eta_{PV,ref}$	Reference efficiency of PV cell	21 %
β	Temperature coefficient of efficiency	0.0041 K^{-1}
T_{PV}	PV cell temperature	℃
$T_{PV,ref}$	Reference PV cell temperature	20 ℃
γ	PV efficiency coefficient	0.12
I_{ref}	Reference sunlight intensity	1000 W/m^2
ε_s	Skyward long-wave emissivity	0.9
σ_s	Stefan-Boltzmann constant	$5.67\cdot10^{-8}$ W/(m^2·K^4)
T_{sky}	Sky temperature, according to ISO 13790	℃
T_{amb}	Ambient temperature	℃

Table 6
Nomenclature and values for PV cell temperature.

Symbol	Description	Units or value
NOCT	Nominal operating cell temperature	48 ℃
R_t	PV to absorber resistance	$0.036\ m^2 \bullet K/W$
R_b	Backside thermal resistance	$10.8\ m^2 \bullet K/W$
T_{back}	Back surface air temperature	℃
$h_{conv,t}$	Top convective loss coefficient	$5.56\ W/(m^2 \bullet K)$

$$\overline{T}_{base} = T_{PVT,fluid} + q_u' z \tag{17}$$

$$T_{fin} = \frac{b}{j} + 2\left(\overline{T}_{base} - \frac{b}{j}\right) \frac{\tanh\left(\frac{m(W_{tube} - d_{tube})}{2}\right)}{m(W_{tube} - d_{tube})} \tag{18}$$

Here, $T_{PVT,fluid}$ is the temperature of the fluid in tube, q_{PVT}' is heat flux in PVT, z is the thermal resistance in the PVT, b is energy balance temperature factor, j is thermal resistance adjust factor, and m is fin efficiency calculation parameter

$T_{PVT,fluid}$, q_{PVT}', z, b, j, and m are calculated as follows:

$$T_{PVT,fluid} = \frac{\left(T_{PVT,entry} + \frac{\varepsilon}{\sigma}\right) \times e^{\frac{n_{tube}\sigma W_{collector}}{\theta \dot{m}_{PVT} C_{p,w}}}}{\frac{W_{collector} n_{tube} \sigma}{\theta \dot{m}_{PVT} C_{p,w}}} - \frac{T_{PVT,entry} + \frac{\varepsilon}{\sigma}}{\frac{W_{collector} n_{tube} \sigma}{\theta \dot{m}_{PVT} C_{p,w}}} - \frac{\varepsilon}{\sigma} \tag{19}$$

$$q_{PVT}' = \frac{q_{PVT}}{n_{tube} W_{collector}} \tag{20}$$

$$z = \frac{1}{h_{PVT,fluid} \pi d_{tube}} + \frac{\lambda_{bond}}{k_{bond} W_{bond}} \tag{21}$$

$$b = S + h_{rad} T_{sky} + h_{conv,t} T_{amb} + \frac{T_{amb}}{R_b F'} \tag{22}$$

$$j = \frac{1}{R_t F'} + \frac{1}{R_b F'} - \frac{1}{R_t} \tag{23}$$

$$m = \left(\frac{F' j}{k_{abs} \lambda_{abs}}\right)^{0.5} \tag{24}$$

where ε is temperature rise efficiency factor, σ is net heat transfer coefficient, n_{tube} is number of identical fluid tubes, θ is thermal transfer effectiveness ratio, $h_{PVT,fluid}$ is the heat transfer coefficient from the fluid to the walls of the fluid channel enclosure, λ_{bond} is thickness of the bond, k_{bond} is the bond thermal conductivity, W_{bond} is width of the bond, k_{abs} is the absorber plate thermal conductivity, and λ_{abs} is thickness of the absorber plate.

The computation of ε, σ, and θ is explained below:

$$\varepsilon = d_{tube} F' b + \frac{nb}{j} \tag{25}$$

$$\sigma = -u - n \tag{26}$$

$$\theta = 1 + z \times u + n \times z \tag{27}$$

Here, n is the conductor heat transfer coefficient, and u is the tube heat transfer product.

Details of the derivations for n and u are as follows:

$$n = 2 k_{abs} \lambda_{abs} m \tanh\left(\frac{m(W_{tube} - d_{tube})}{2}\right) \tag{28}$$

$$u = d_{tube} F' \left(h_{rad} + h_{conv,t} + \frac{1}{R_b F'}\right) \tag{29}$$

Finally, the PVT fluid exit temperature and solar heat collection net efficiency (η_{th}) are calculated as follows:

$$T_{PVT,exit} = \left(T_{PVT,entry} + \frac{\varepsilon}{\sigma}\right) e^{\frac{n_{tube}\sigma W_{collector}}{\theta \dot{m}_{PVT} C_{p,w}}} - \frac{\varepsilon}{\sigma} \tag{30}$$

$$\eta_{th} = \frac{q_{PVT}}{A_{collector} I_{module} + P_{pump,PVT}} \tag{31}$$

Here, $P_{pump,PVT}$ is the power consumption of the PVT circulator pump.

2.2.3. Heat pump model

In this study, we employed both polynomial regression and Multilayer Perceptron (MLP) models to predict the outlet condition of the refrigerant in a heat pump system, considering the inlet conditions of its primary components such as the evaporator, compressor, condenser, and expander. The choice between these models was guided by the nature of the relationship between inlet and outlet states of the refrigerant: linear relationships were analyzed using regression models, while nonlinear ones were explored through the MLP model. Despite the higher computational demands of MLP models, their effectiveness in capturing complex nonlinear relationships with low correlation between inputs and outputs justifies their use.

The MLP model used in this research underwent z-score normalization for data preprocessing and was split into validation and test sets. Its architecture includes an output layer with a single neuron to improve prediction accuracy and incorporates weight initialization to avoid gradient loss (Salimans and Kingma, 2016). During forward propagation (Park and Lek, 2016), mean square error (MSE) was the loss function used (Murphy, 1988), and in backward propagation (Hecht-Nielsen, 1992), gradient calculation employed Leaky ReLU (Sharma et al., 2020), with L2 regularization to prevent overfitting (Ying, 2019). An early stopping mechanism halted training after 500 epochs without improvement in validation error. Hyperparameter optimization, including learning rates and regularization strengths, was determined through grid search (Bergstra and Bengio, 2012).

Our research focused on implementing a shell-and-coil configuration to effectively transfer thermal energy from a TES system to a heat pump. Water circulates on the shell side directly connected to the TES, and refrigerant circulates on the coil side connected to the compressor. Predictions for initial refrigerant pressure in the coil were based on shell side temperature, utilizing operational data from the scroll compressor. This was key in determining the refrigerant flow rate and the compressor's efficiency.

The efficiency (η_{comp}) of the scroll compressor was calculated using the suction temperature ($T_{suction}$) and evaporation pressure (P_{evap}) as follows:

$$\eta_{comp} = -0.006379 T_{suction} + 0.0012187 P_{evap} \tag{32}$$

Predictions based on short operational intervals may be less precise, but these inaccuracies are not expected to significantly affect the heat pump's overall performance. Therefore, an extensive analysis of this aspect is not included in this study.

In heat pump systems, the shell-and-coil model used the finite volume method, which, despite longer computation times due to the division of length into numerous elements, offers precise temperature and pressure calculations for each element. Advances in computing power and optimized element spacing help overcome computational limitations. Fig. 5 shows the counterflow shell-and-coil numerical model using this method.

When the circulator pump of the HP is operational, the heat transfer rejected from the coil side by the water in the shell (q_{shell}) is calculated using the total thermal resistance to heat transfer between the two fluids in the shell and coil ($R_{shellandcoil}$), combined with the temperature differential between these fluids on the shell and coil sides. This is expressed by the following equation:

$$q_{shell,i} = \frac{1}{R_{shellandcoil,i}} \left(T_{shell,i} - T_{coil,i}\right) \tag{33}$$

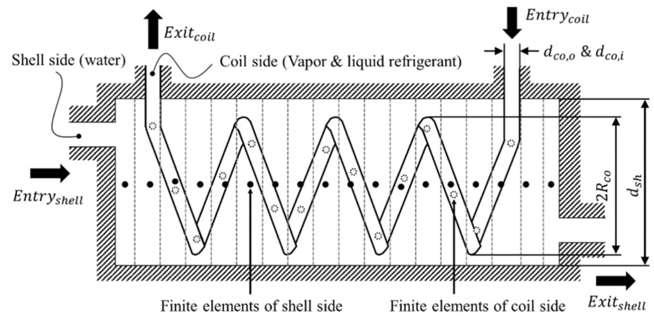

Fig. 5. Numerical shell-and-coil model based on finite volume method. $d_{co,o}$: coil outside diameter; $d_{co,i}$: coil inside diameter; R_{co}: coil radius; and d_{sh}: shell diameter.

Here, i is the finite element number of the shell and coil, T_{shell} is temperature of water flowing from shell side, and T_{coil} is temperature of refrigerant flowing from coil side.

The thermal resistance is defined as:

$$R_{shellandcoil} = \frac{1}{4\pi d_{coil,i} L_{coil} h_{coil}} + \frac{\ln\left(d_{coil,o}/d_{coil,i}\right)}{2\pi k_{coil} L_{coil}} + \frac{1}{4\pi d_{coil,o} L_{coil} h_{shell}} \tag{34}$$

In these equations, L_{coil} represents the coil length differentiated by finite elements, h_{coil} is the convective heat transfer coefficient on the coil side, k_{coil} is the coil's thermal conductivity, and h_{shell} is the convective heat transfer coefficient on the shell side. The convective heat transfer coefficient due to fluid flow used the Nusselt number calculation formula proposed in related research (Salimpour, 2009).

Regarding the dynamic viscosity, kinematic viscosity, and thermal expansion of water, these were predicted using a polynomial regression model. The performance of the heat pump model used in this study should be considered for outdoor and indoor plate-fin-and-tube heat exchangers (Taler et al., 2020). However, because the purpose of this study is to quantitatively evaluate the performance of the integrated PVT-HP system by flow rate of the circulator pump, the system was designed so that the dryness exceeds 1.0 through heat transfer of the shell and coil alone. Therefore, HP's coefficient of performance (COP) is calculated as follows:

$$\mathrm{COP} = \frac{\left(h_{co,exit} - h_{co,entry} + W_{comp}/\dot{m}_{coil}\right)}{W_{comp}/\dot{m}_{coil}} \tag{35}$$

$$\dot{m}_{coil} = \frac{W_{comp}\eta_{comp}}{h_{discharge} - h_{suction}} \tag{36}$$

Here, $h_{co,exit}$ is coil fluid exit enthalpy, $h_{co,entry}$ is coil fluid entry enthalpy, W_{comp} is power rating of compressor, m_{coil} is coil fluid flow rate, $h_{discharge}$ is enthalpy of compressor discharge, and $h_{suction}$ is enthalpy of compressor suction.

2.2.4. Thermal energy storage model

Accurate heat transfer analysis in the TES model necessitates a precise definition of control parameters for each node. Centroid locations and heat transfer calculations for each control volume in the TES model were performed using differential methods grounded in the Cartesian coordinate system. The dimensions of each finite element in the x- and y-axes are dynamically set based on the number of central angles specified by the user. Fig. 6 displays the three-dimensional configuration of the TES numerical model.

In the finite element model, heat transfer via conduction, convection, and mixing between nodes must adhere to principles of momentum balance and energy balance (Reddy and Gartling, 2010).

Heat transfer processes, both through conduction as per Fourier's law and mixing in accordance with the Navier-Stokes equation, are quantified as follows (Kothandaraman, 2006):

$$q_{stream} = \rho_w C_{p,w}\left[\left(u\frac{\partial T}{\partial x} + T\frac{\partial u}{\partial x}\right) + \left(v\frac{\partial T}{\partial y} + T\frac{\partial v}{\partial y}\right) + \left(w\frac{\partial T}{\partial z} + T\frac{\partial w}{\partial z}\right)\right] dxdydz \tag{37}$$

$$q_{cond} = \frac{\partial}{\partial x}\left(k_w\frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k_w\frac{\partial T}{\partial y}\right) + \frac{\partial}{\partial z}\left(k_w\frac{\partial T}{\partial z}\right) \tag{38}$$

Here, u, v, and w are the velocity components in the x, y, and z directions, respectively, ∂T represents the temperature gradients, and heat transfer as the outermost nodes includes heat exchange due to the ambient temperature.

The method of calculating heat transfer by internodal convection in

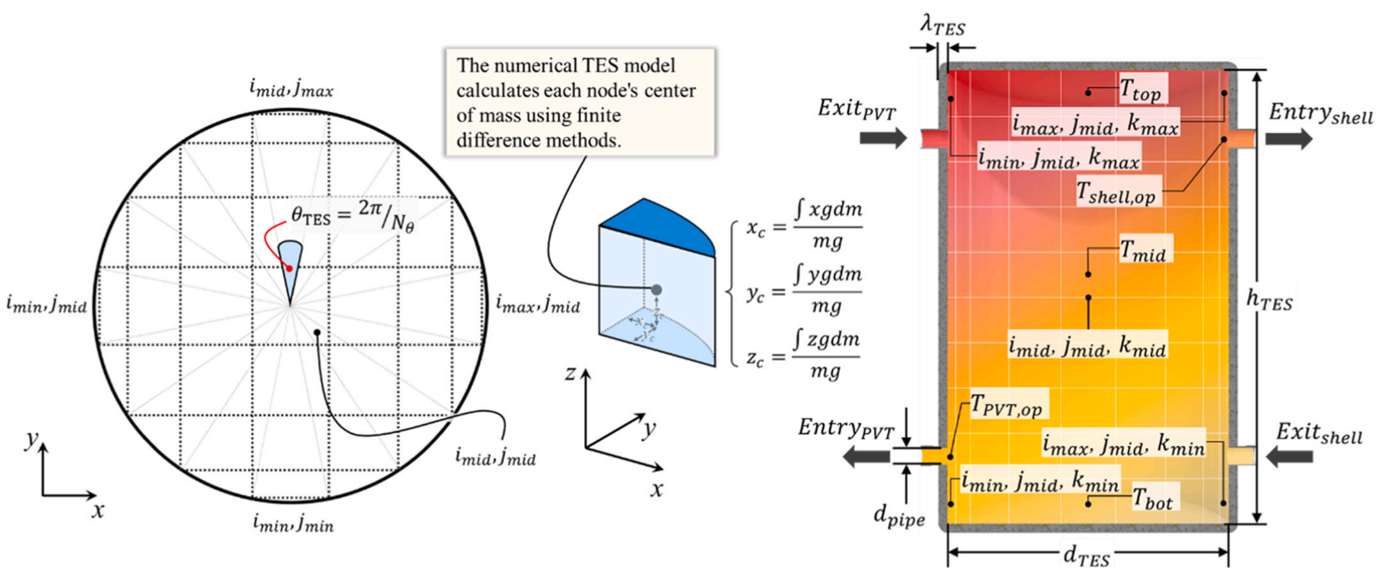

Fig. 6. Three-dimensional TES numerical model: design and temperature nodes. θ_{TES}: central angle size, x_c: x-axis center of mass of finite element, y_c: y-axis center of mass of finite element, z_c: z-axis center of mass of finite element, d_{pipe}: pipe diameter; d_{TES}: TES diameter; T_{top}: top temperature of TES; T_{mid}: mid temperature of TES; T_{bot}: bottom temperature of TES; $T_{PVT,op}$: observation point temperature for operation of PVT system, $T_{shell,op}$: observation point temperature for operation of heat pump system.

finite elements is determined using the following equation:

$$q_{cnv} = h_{cnv} A_{adj} \Delta T_{adj} \tag{39}$$

where h_{cnv} is the convective heat transfer coefficient, A_{adj} is area of contact between nodes, and ΔT_{adj} is temperature difference between nodes in contact with each other.

It should be noted that the convective heat transfer coefficient for each node is determined by whether or not the circulator pump is operating. When the circulator pump operates, finite element convective heat transfer is assumed to occur when a heat flux is applied to the moving plate. The forced convective heat transfer coefficient due to the operation of the circulator pump must be calculated separately by dividing the direction of flow based on the Cartesian coordinate system (Bergman, 2011).

Conversely, when the circulator pump is not operating, it is assumed that finite element convective heat transfer occurs when the nodes form fixed vertical and horizontal planes. The natural convective heat transfer has been the subject of much experimental and numerical research, and numerous correlations for *Nu* exist. In general, natural convection is applicable only to a narrow range of Pr, *Ra*, and aspect ratio. *Nu* for the vertical plane between nodes referred to the equations proposed by Catton (1978) and Berkovsky & Poleviko (1977) (Yunus and Afshin, 2015).

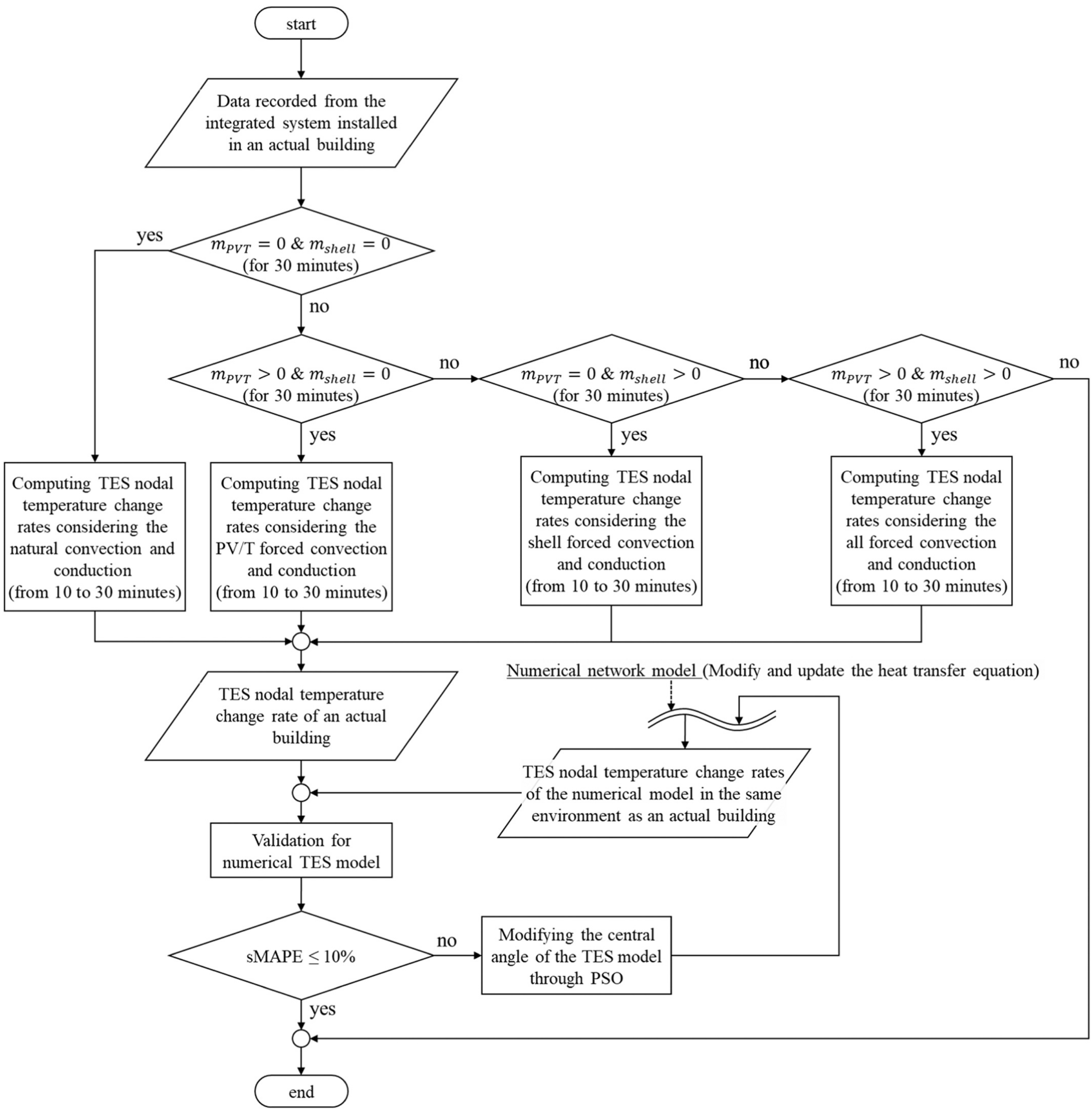

Fig. 7. Flowchart depicting the validation process for the numerical TES model.

2.2.5. Experimental validation

The accuracy of heat collection and power generation of the PVT numerical model and the accuracy of the heat production and COP of the HP numerical model were confirmed in previous studies. So, validation of the PVT and HP numerical models were not conducted separately in this study (Chae et al., 2023a).

Meanwhile, to validate the developed TES model, the predicted value of the numerical model was compared with the actual value measured at the experimental site. The subject of comparison is the temperature change rate for the intermediate node of TES according to flow rate, and the verification method used the symmetric mean absolute percentage error (sMAPE), which is widely used for model verification in the energy simulation field (Goodwin and Lawton, 1999). In order to completely collect actual temperature change rate data, when the fluid flow to be observed lasted for 30 minutes, the data collected for the first 10 minutes was removed, and data for the remaining 20 minutes was collected. The tolerance range of *sMAPE* was set at 10 %, which is considered to have excellent accuracy in many prediction scenarios. If the tolerance range is exceeded, the heat transfer equation should be modified and supplemented by examining related research data, or through Particle Swarm Optimization (PSO) (Van den Bergh and Engelbrecht, 2004). The numerical model was calibrated by changing the size of the finite elements of the TES. The flowchart below shows the validation and calibration process of the numerical TES model. Figs. 7 and 8

The PSO constraints were defined based on the fulfillment of momentum and energy balance and accuracy with actual experimental results. The PSO process encompasses several phases: initialization, evaluation, and updating of velocity and position. Initially, a swarm of random solutions, termed 'particles,' is generated. Each particle symbolizes a specific central angle size in the TES model. During the evaluation phase, the performance of each particle is assessed.

The update phase involves recalculating each particle's velocity and position. This calculation is influenced by the particle's optimal position and the optimal positions of neighboring particles, proceeding iteratively until a predefined maximum number of iterations is reached.

The velocity (v_i^t) and position ($\theta_{TES,i}^t$) of each particle are determined using the following equations:

$$v_i^{t+1} = w \cdot v_i^t + c_1 \cdot r \cdot \left(p_{best,i} - \theta_{TES,i}^t\right) + c_2 \cdot r \cdot \left(g_{best} - \theta_{TES,i}^t\right) \tag{40}$$

$$\theta_{TES,i}^{t+1} = \theta_{TES,i}^t + v_i^{t+1} \tag{41}$$

In these equations, v_i^{t+1} is the new velocity for adjusting the central angle size, w is the inertia weight, controlling the influence of the previous velocity, c_1 denotes the cognitive coefficient, r denotes a random number, $p_{best,i}$ is the best central angle size found so far by the i-th particle, θ_i^t represents the central angle size of the TES for the i-th particle, c_2 is the social coefficients, and g_{best} is the best central angle size found so far by any particle in the swarm. The cognitive and social accelerations, as well as the initial and final inertia weights, were set at 1.5, 2, 0.8, and 0.4, respectively, through direct tuning. Consequently, the optimized central angle size was calculated to be 0.125π radians. By comparing the optimized TES model with the results measured in an actual building, the sMAPE for the temperature change rate of the intermediate node was calculated to be approximately 1.47 % when all circulator pumps were stopped and about 3.29 % when more than one circulator pump was operating. It should be noted that the results are limited by the fact that the actual measured data is based on buildings that are currently in operation, so the number of measured data points for the experiment is not large, and there are limitations due to the lack of temperature observation points. However, to ensure sophisticated verification of the numerical model in a limited environment, not only the temperature of each node but also the influence of external temperature and inlet/outlet temperature were considered, thereby increasing the sophistication of the verification process.

This study also revealed that the average error rate of the one-dimensional TES model was 35 %, while the 3D TES model showed an average error rate of 2.05 %. These results suggest that the 3D TES model provides more accurate predictions.

The equations below represent the sMAPE, temperature change rate of the TES central node, respectively.

$$\text{sMAPE} = \frac{100}{n}\sum_{i=1}^{n}\frac{\left|\vartheta_{pred} - \vartheta_{act}\right|}{\left(\left|\vartheta_{pred}\right| + \left|\vartheta_{act}\right|\right)/2} \tag{42}$$

$$\vartheta = \frac{\left|\vartheta_t - \vartheta_{t-20}\right|}{\vartheta_{t-20}} \times 100 \tag{43}$$

Here, ϑ_{pred} is the predicted temperature change rate, ϑ_{act} is the actual temperature change rate, and t is time sequence.

The figure below shows the actual measured results and the verification results of the optimized TES model.

3. Results

The performance evaluation of the integrated PVT-HP system,

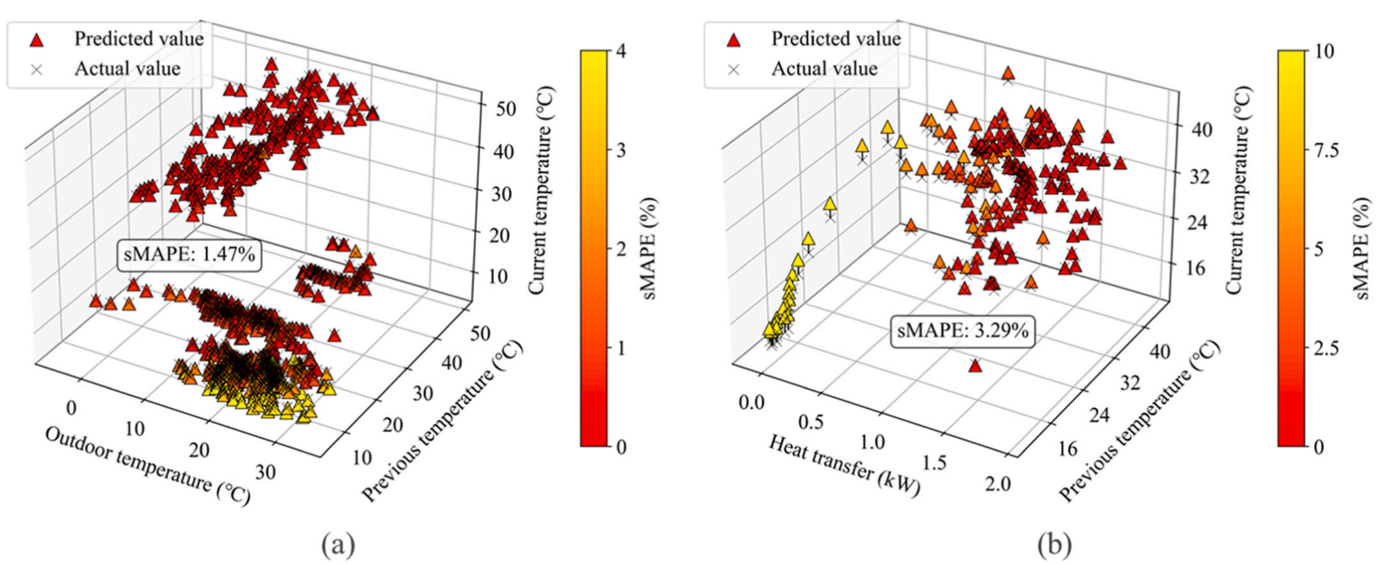

Fig. 8. TES model validation: comparison at natural and forced convection outcomes: (a) natural convection and (b) forced convection.

alongside the temperature gradient analysis of the TES as influenced by flow dynamics, was predicated on the heat storage capabilities of the PVT and the heating functionality of the HP. The performance analysis of the system was conducted by selecting December 14th to December 16th as representative days. Heating operation is performed when $T_{shell,op}$ is higher than the set temperature ($T_{shell,sp}$), $T_{shell,sp}$ is 30 ℃, and deadband ($T_{shell,db}$) is 5 ℃. The initial heat storage operation is performed when the sum of the deadband ($T_{PVT,db}$) of $T_{PVT,op}$ and PVT is greater than $T_{PVT,fluid}$, and the heat storage operation is set to continue only when q_{PVT} is more than 10 times the power consumption of the circulator pump. The flow rates of PVT and HP are the same, and this is to prevent failure of the circulator pump directly connected to the nozzle-less TES. Fig. 9 shows the heating and heat storage operation control algorithm of the integrated PVT-HP system.

3.1. Photovoltaic/thermal system

As a result of numerical analysis of the performance of the PVT system by flow rate during a representative day, q_{PVT} was 8.6 % higher and P_{PV} was 1 % lower when operated at the minimum flow rate compared to the maximum flow rate. Within the analysis range, as the flow rate increased by 10 LPM, η_{th} decreased by 9.1 %, and there was little change in η_{PV}. Fig. 10 shows the results of numerical analysis of PVT system performance by flow rate during December 16.

3.2. Heat pump system

As a result of numerical analysis of the performance of the HP system by flow rate on the shell side during representative days, q_{shell} was 7.8 % higher when operated at the minimum flow rate compared to the maximum flow rate. Within the analysis range, as the flow rate decreased by 10 LPM, COP decreased by 1.1 % and a log mean temperature difference (LMTD) increased by 1.2 %. Fig. 11 shows the numerical analysis results of the HP system by flow rate during December 16.

The increase in LMTD due to the decrease in flow rate caused a decrease in refrigerant enthalpy and refrigerant entropy in compressor suction. This change reduced the refrigerant pressure difference between discharge and suction, increasing compression efficiency. Within the analysis range, as the flow rate on the shell side decreased by 10 LPM, the outlet enthalpy of the coil, η_{comp}, and $\dot{m}_{coil}$ decreased by 0.4 %, 1.1 %, and 0.3 %, respectively.

3.3. The integrated PVT-HP system

This section presents the results of numerical analysis of temperature changes at the top/middle/bottom of the TES and heat transfer on the heat source and load sides during a representative day. Within the scope of analysis, the difference between T_{top} and T_{bot} increased by 33.5 % as the flow rate decreased by 10 LPM, and the operating time of the PVT system and HP system increased by 3.5 % and 3.9 %, respectively. Fig. 12 shows the temperature gradient, operating time, and heat transfer of TES by flow rate during December 16.

Taking the results together, several key factors significantly influence the performance of each component in the integrated PVT-HP system:

- Flow rate: The flow rate of circulating water affects heat transfer efficiency and overall system performance. Proper management optimizes temperature gradients and enhances the system's heat storage capacity and the COP of the heat pump.
- Temperature gradients: Maintaining optimal temperature gradients within the TES is crucial for efficient energy storage and retrieval. A

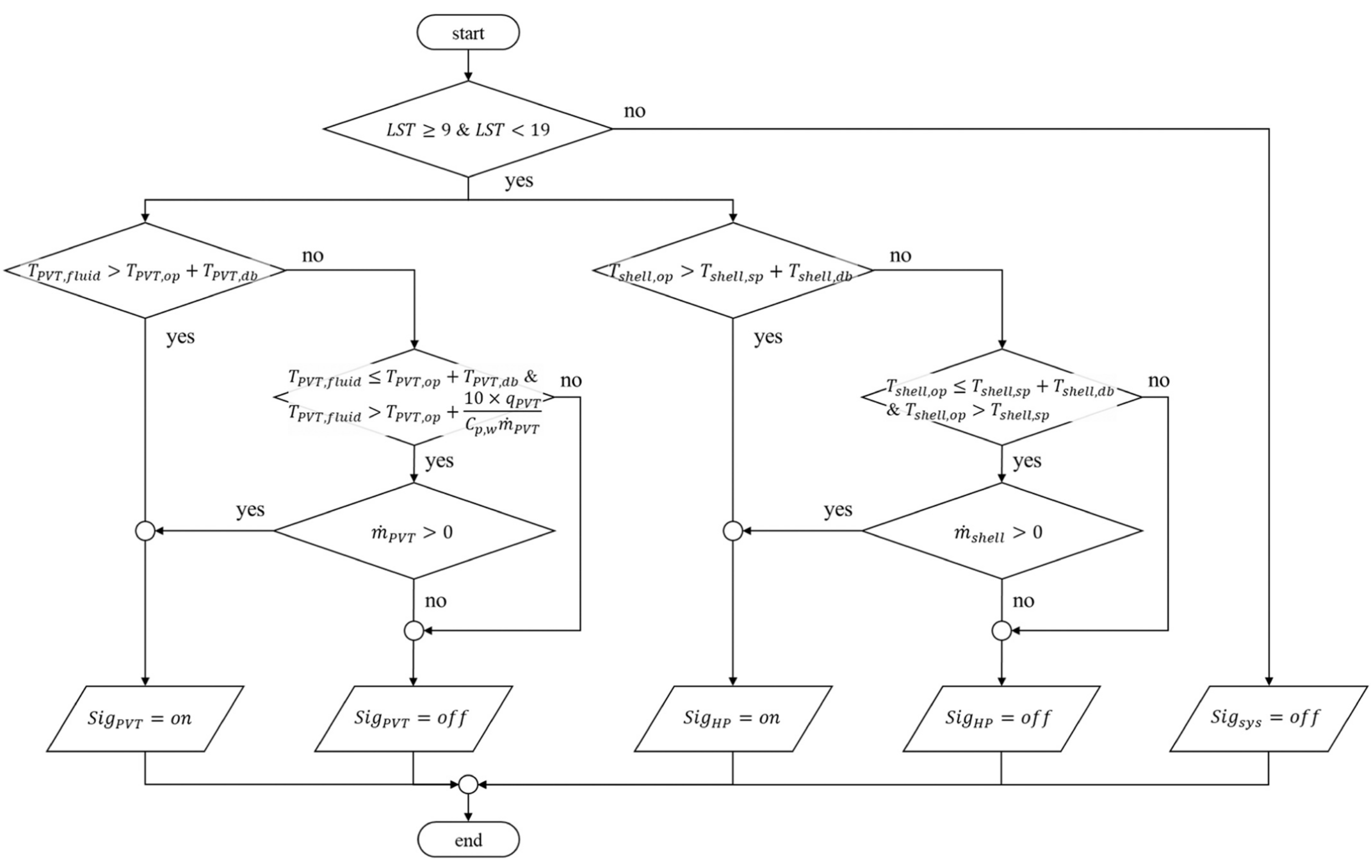

Fig. 9. Operational control algorithm for heat storage and heating in the PVT-HP system.

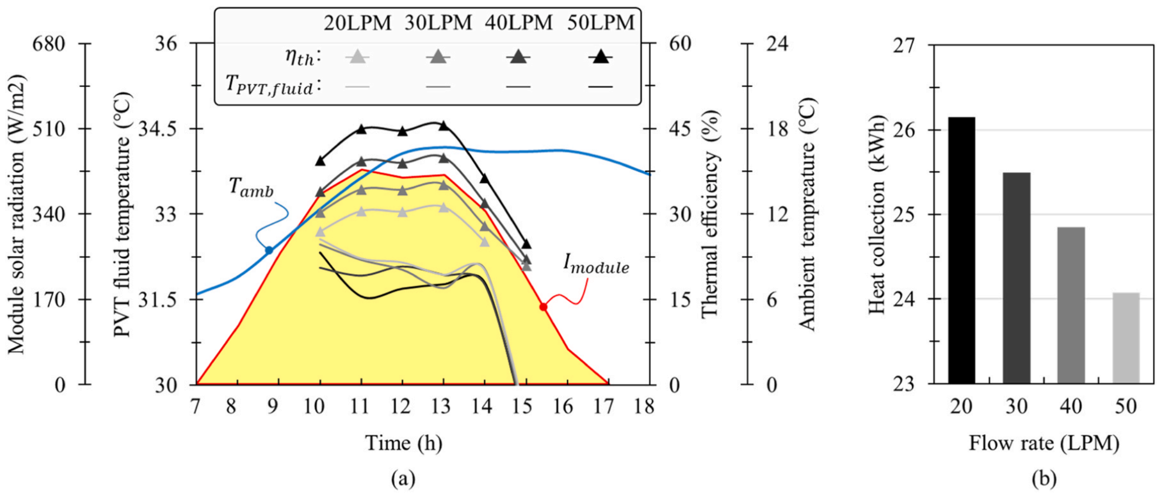

Fig. 10. PVT system performance variation with flow rate changes: (a) PVT efficiency over time, and (b) Heat collection amount.

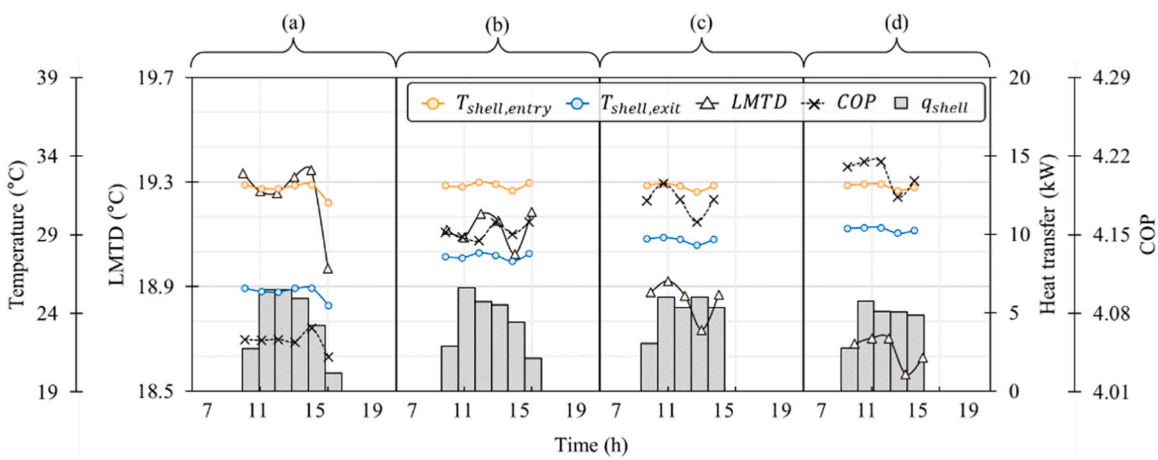

Fig. 11. Flow rate impact on HP system performance: (a) 20 LPM, (b) 30 LPM, (c) 40 LPM, and (d) 50 LPM.

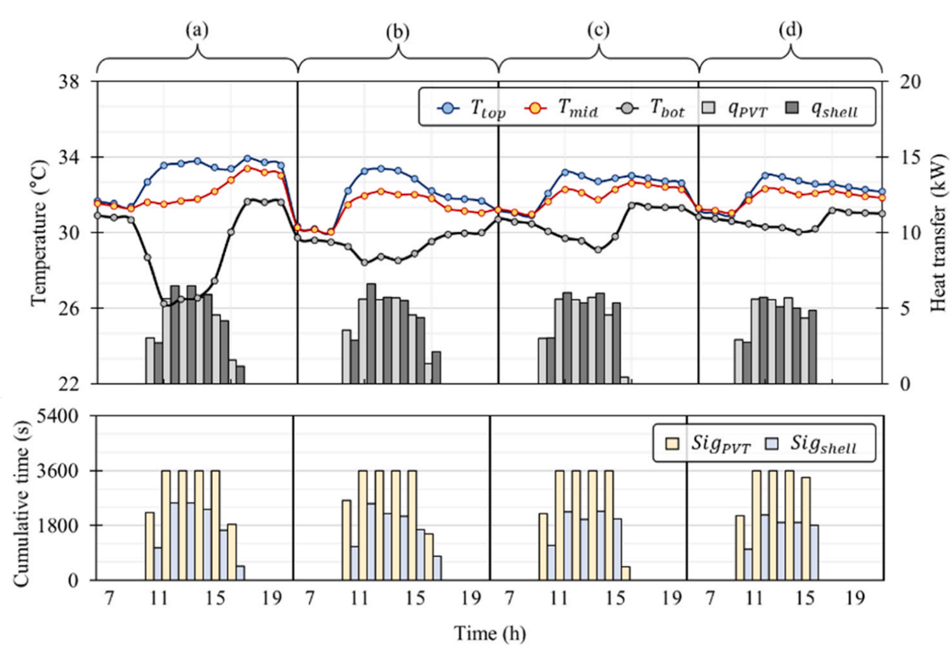

Fig. 12. Integrated PVT-HP system performance analysis by flow rate: (a) 20 LPM, (b) 30 LPM, (c) 40 LPM, and (d) 50 LPM.

well-maintained gradient ensures effective thermal stratification, allowing efficient extraction and use of stored heat.

- Solar radiation: The amount of solar radiation incident on the PVT modules determines the system's energy generation and thermal collection efficiency. Accurate estimation and optimization of incident solar radiation maximize system output.
- Heat exchanger efficiency: The performance of heat exchangers, including the shell-and-coil configuration, directly influences the system's efficiency. Efficient thermal exchange ensures minimal thermal losses and effective heat transfer.
- Environmental conditions: Surrounding environmental conditions, such as ambient temperature and humidity, impact the PVT-HP system's performance. Incorporating real-time environmental data into control algorithms helps maintain optimal performance under varying conditions.

By understanding and optimizing these factors, the integrated PVT-HP system's performance can be significantly improved, leading to better energy efficiency, higher reliability, and enhanced sustainability of building energy systems.

4. Discussion

4.1. Correlation between flow rate and vertical temperature gradient

The TES model in this study is three-dimensional, analyzing fluid flow in both vertical and horizontal directions. Typically, increasing flow rate reduces the vertical temperature gradient, but the model's complexity requires review to ensure normal vertical gradients. Data on temperature differences between upper and lower central nodes of the TES under various flow rates and heat transfer conditions were collected, including scenarios with only the PVT system operational and others with both the PVT and HP active. A significant finding is a Pearson correlation coefficient of −0.98 between flow rate and temperature difference, indicating a strong inverse relationship. Fig. 13 shows heat transfer and temperature gradients at different flow rates, with total heat transfer being the sum of q_{PVT} and q_{shell}. Subfigures (a) to (d) depict PVT-only operation at flow rates of 20, 30, 40, and 50 LPM, while (e) to (h) show combined operation at the same flow rates, with outliers removed for clarity.

Fig. 13 highlights that flow rate changes directly impact the temperature gradient within the TES system. The data suggests that an increased flow rate on the PVT side raises the z-axis momentum vector of the fluid exiting the PVT, causing a rapid increase in $T_{PVT,op}$. This narrows the temperature differential between $T_{PVT,op}$ and $T_{PVT,fluid}$, reducing heat storage capacity and operations. Higher flow rates also decrease η_{PV} and η_{th} due to increased average $T_{PVT,fluid}$ and T_{PV} from fewer heat storage operations. Moreover, an increased flow rate enhances average COP during operation due to higher $T_{shell,etnry}$, but reduces the temperature gradient in the TES, limiting $T_{shell,op}$ and resulting in fewer heating operations and lower accumulated q_{shell}. Strategically controlling the circulator pump flow rate according to environmental conditions could improve overall system performance.

4.2. Impact of flow rate adjustments on thermal storage and transfer

Strategic adjustments in the flow rate of circulating water significantly impact the thermal storage and transfer capabilities of the integrated PVT-HP system:

- Temperature gradients within TES: Lower flow rates increase the temperature difference between the top and bottom of the TES by up to 33.5 %, enhancing thermal stratification and storage efficiency.
- Heat transfer efficiency: Adjusting the flow rate affects the convective heat transfer coefficient. A decrease in flow rate by 10 LPM can

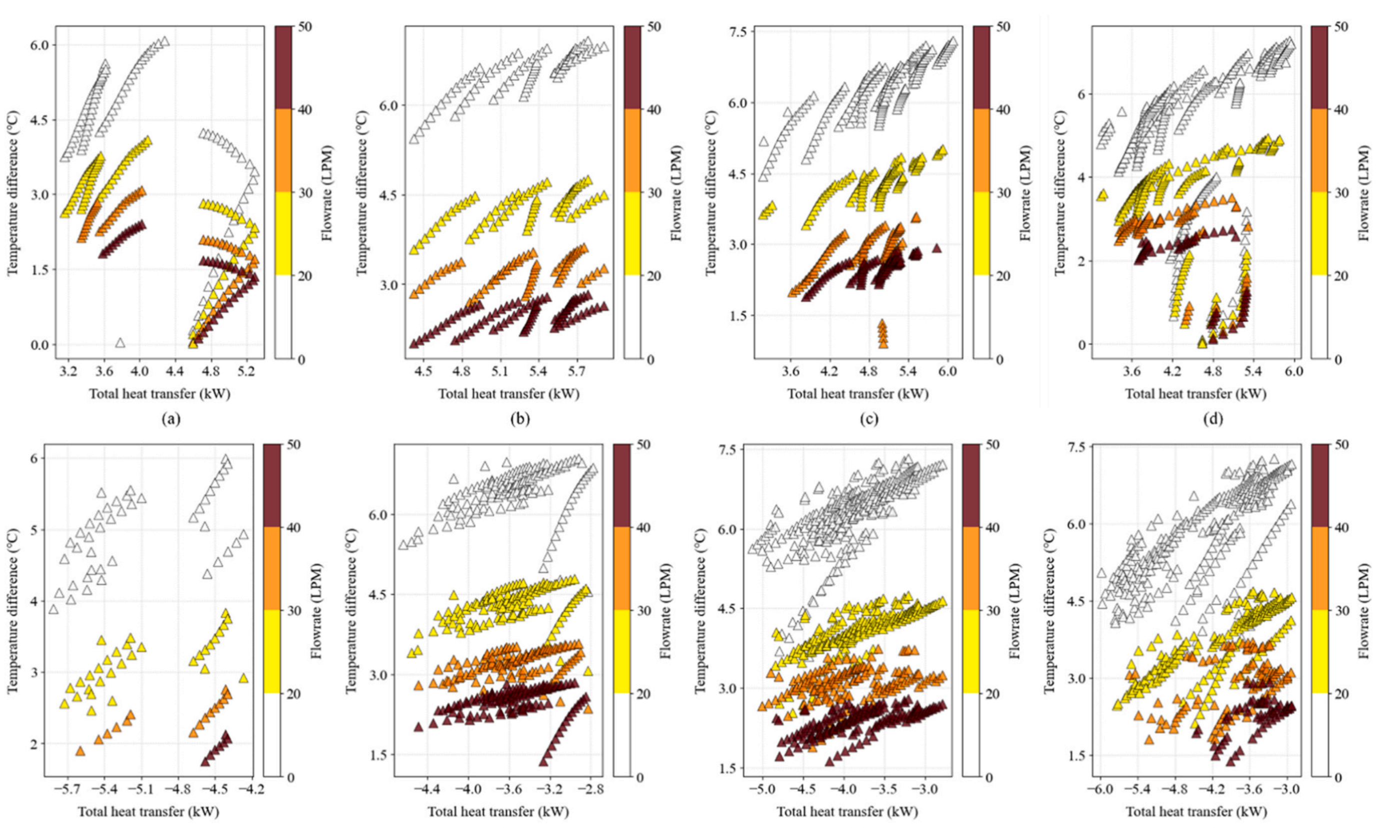

Fig. 13. Impact of flow rate on temperature gradient in TES: top vs. bottom.

lead to a 1.1 % decrease in the heat pump's COP and a 1.2 % increase in the LMTD.
- System's coefficient of performance: Optimal flow rates maintain suitable temperature differentials across the heat pump, maximizing its COP. For instance, a flow rate decrease by 10 LPM can result in a 1.1 % reduction in COP.
- Thermal energy extraction and utilization: Reducing the flow rate during low demand periods conserves thermal energy within the TES. Conversely, increasing the flow rate during peak demand enhances the delivery of stored thermal energy, ensuring consistent heating or cooling.

By strategically adjusting the flow rate based on real-time data, the PVT-HP system can achieve optimal thermal storage and transfer efficiency, balancing energy supply with demand and improving overall performance.

5. Conclusion

This study presented an advanced numerical analysis of an integrated PVT-HP system, underscored by a novel three-dimensional TES model. The key findings highlight the substantial enhancements in system performance attributable to strategic flow rate adjustments and the application of a 3D TES model, which are as follows:

- The PVT system's heat collection (q_{PVT}) increased by 8.6 % at the minimum flow rate compared to the maximum flow rate. The power generation (P_{PV}) showed a minimal decrease of 1 % under the same conditions. The thermal efficiency (η_{th}) decreased by 9.1 % with each 10 LPM increase in flow rate, while PV efficiency (η_{PV}) remained relatively unchanged.
- The heat supply (q_{shell}) increased by 7.8 % at the minimum flow rate compared to the maximum flow rate. The coefficient of performance (COP) of the heat pump decreased by 1.1 % with each 10 LPM decrease in flow rate, and the log mean temperature difference (LMTD) increased by 1.2 %.
- The difference in temperature between the top and bottom of the TES increased by 33.5 % with each 10 LPM decrease in flow rate. The operating time for the PVT and HP systems increased by 3.5 % and 3.9 %, respectively, with lower flow rates.
- The 3D TES model demonstrated superior accuracy in predicting temperature distributions compared to conventional one-dimensional models, especially under dynamically changing conditions. The Pearson correlation coefficient of −0.98 between flow rate and temperature gradient within the TES system highlights a strong inverse relationship, crucial for optimizing system performance.

The findings of this research underscore the critical role of strategic flow rate control in enhancing the performance of integrated PVT-HP systems. The three-dimensional TES model not only improves the accuracy of thermal predictions but also facilitates more efficient control logic development and system capacity determination. Future studies should focus on simplifying the 3D TES model for practical applications and exploring remote control technologies to overcome computational limitations. Additionally, integrating high-performance optimal control technologies, including artificial intelligence, could further enhance system efficiency and sustainability.

Efforts should be directed towards simplifying the TES model to ensure its practical application in real-world scenarios, particularly in micro control units (MCUs) installed in general control devices. Furthermore, exploring the integration of artificial intelligence technologies to develop high-performance optimal control strategies that can dynamically adjust system parameters for optimal performance could provide significant advancements. This study provides significant insights into the integrated performance evaluation of PVT-HP systems and presents a novel methodology for efficient energy system management. These findings contribute to the advancement of sustainable energy technologies and offer a robust foundation for future research in this domain.

CRediT authorship contribution statement

Soowon Chae: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Data curation, Conceptualization. **Yujin Nam:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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